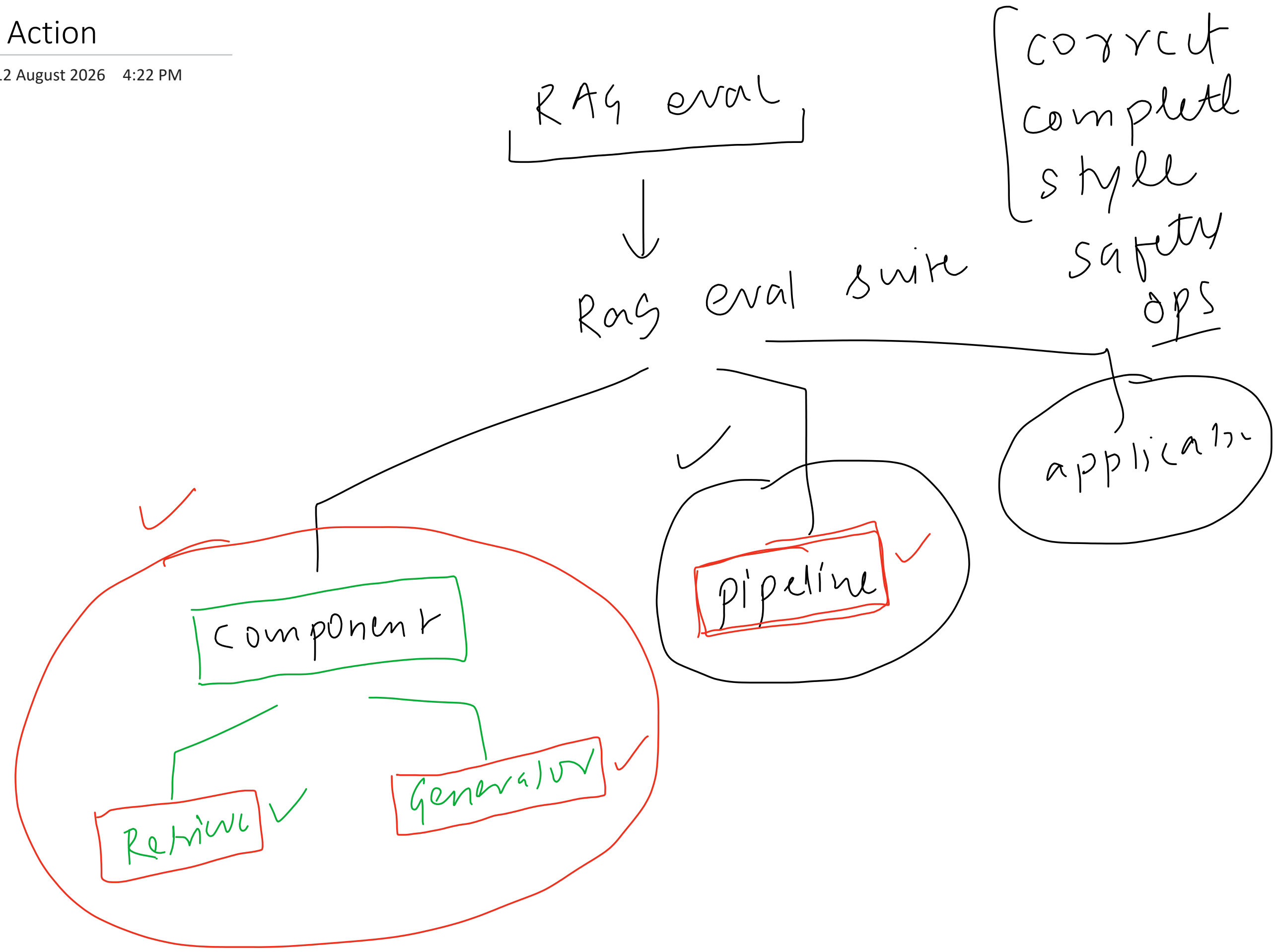
Plan of Action

Wednesday, 12 August 2026 4:22 PM



Creating the Generator

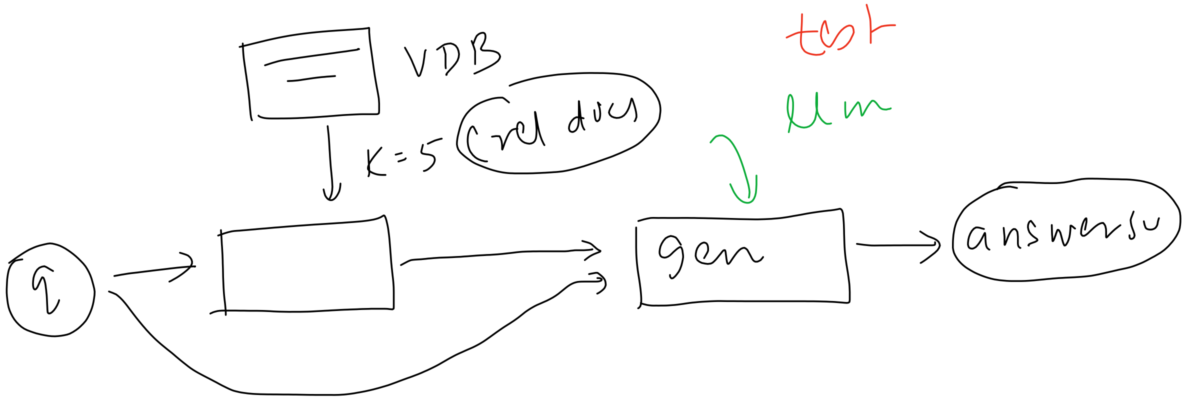
Wednesday, 12 August 2026 4:22 PM

[Generator]

rag app compm

↓

test



Generator Failure Modes

Wednesday, 12 August 2026 4:23 PM

unfaithful respons

X faithfulness → co → corr

Query

> Does the CampusX AI Engineering program include live classes?

Retrieved context

> The AI Engineering program includes recorded lessons, coding assignments, projects, and weekly doubt-solving sessions.

→ context

Yes. The program includes two live classes every week along with weekly doubt-solving sessions.

→ answer

How to calculate Faithfulness & Answer Relevance

Wednesday, 12 August 2026 4:23 PM

olation evaluat

[dataset] → generator → ans

question

golden context

1 what is rag triad

what are online evals

3/5

3/5

LLM

Query: what does it mean for a benchmark to get saturated?

Context (what the generator was given)
Saturation = scores cluster high & close together (95%, 94%, 92%)
→ can't tell models apart → benchmark no longer useful

Generated answer (this is what we score)
"A benchmark gets saturated when it can no longer distinguish between models. This happens once models all score very high on it, and the models show a wide range of scores across the benchmark."

↓ decompose into claim

Claim 1 — a saturated benchmark no longer distinguishes mode
context says exactly this ("can't differentiate the models")

Claim 2 — it happens once models all score very high
context supports this (95% / 94% / 92% clustering)

Claim 3 — models show a wide range of scores
CONTRADICTS context — saturation is clustering, not spreading

↓

supported ÷ total = 2 ÷ 3
Faithfulness = 0.67

Retri — gen

powerful judge → 40 /

Query: what does it mean for a benchmark to get saturated?
relevance checks every statement against THIS — not against the context

Generated answer (this is what we score)
"A benchmark gets saturated when model scores cluster high and close together, so it can no longer tell models apart. Separately, benchmark contamination happens when test data leaks into training data."

↓ split into statements

Statement 1 — saturation is when scores cluster high & close
directly answers the query

Statement 2 — so the benchmark can't tell models apart
directly answers the query

Statement 3 — benchmark contamination is test data leaking
OFF-TOPIC — true & in the context, but doesn't answer "saturation"

↓

relevant ÷ total = 2 ÷ 3
Answer relevance = 0.67

q / context → generator → answer

golden

q — LLM

answe

claims

C1 ✓

C2 ✓

C5 ×

$\frac{2}{3}$

Faithfulness Golden Dataset

Wednesday, 12 August 2026 4:23 PM

15 dataset

question

golden_context → add json

vector_db

chunks

export

claude

Deepeval synthe

862 chunks

export

json file

DeepEval Code

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faith → 0.91 → 0.78

q

c

LLM
instruct

answer

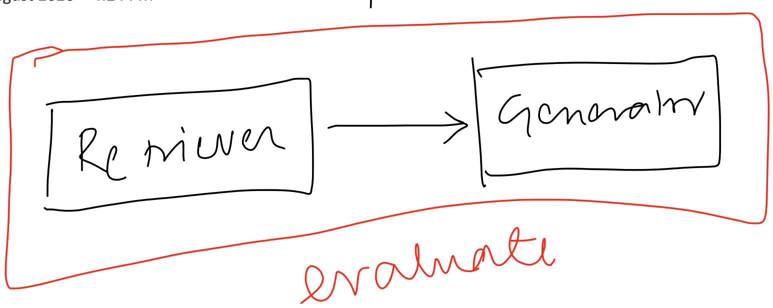
retriever → chunk size
embedding
reranker

generator
better model
system prompt

Building the RAG Pipeline

Wednesday, 12 August 2026   4:24 PM

RAG pipeline



✓ step 1 — build this pipeline

→ step 2 — evaluate

RAG Triad

Wednesday, 12 August 2026 4:24 PM

Golden
gen

question

answer relevancy

contextual relevancy

reference free

answer

context

faithfulness

$\frac{10}{15} =$

k=5

K=2

15 claims

q

retriever

context

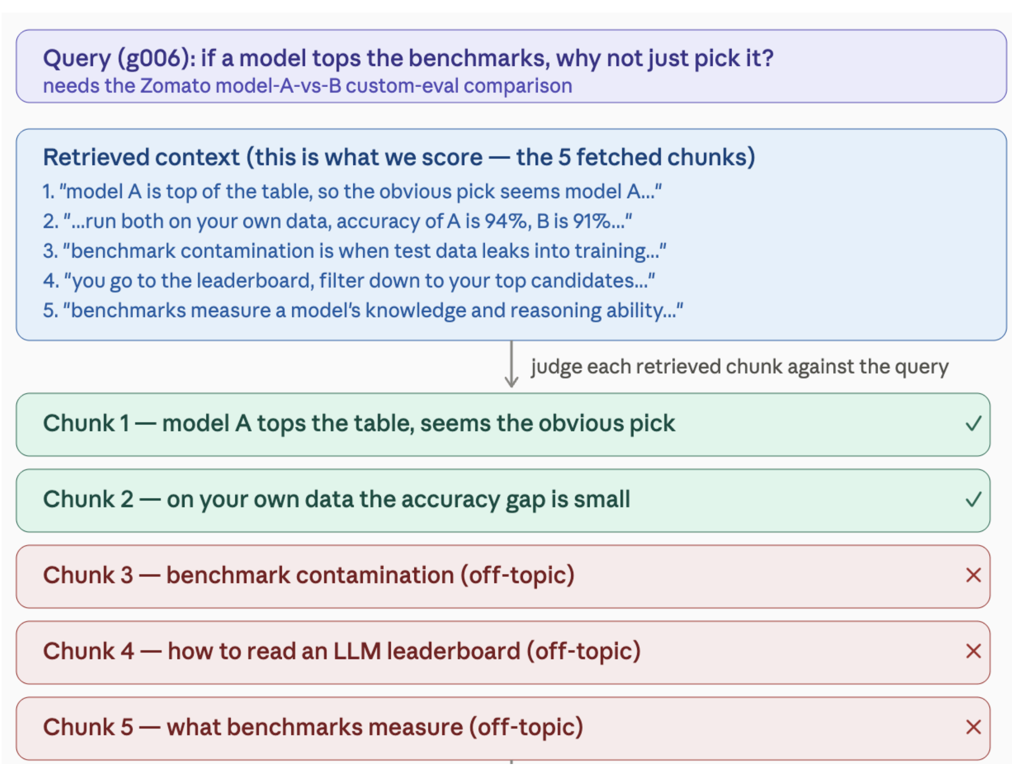
LLM → C1 →

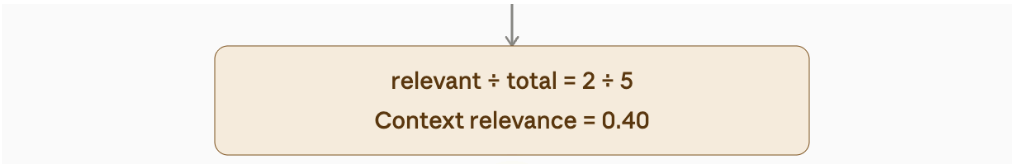
Claim 1 ✓

Claim 2 ✓

claim 3 ✗

c - - - ✓




relevant ÷ total = 2 ÷ 5
Context relevance = 0.40

DeepEval Code

Wednesday, 12 August 2026 4:24 PM

faith → 0.93

ans rel → 0.86

contextual relevancy → 0.42

retriever

recall – 99

precision – 89

42%

2/5

5 claims

K = 5

✓ C1 → 5 lines → claims

✓ C2 → 5 cla. 1-n

✓ C3

✓ C4

× C5